

Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.11)

Metric

$$ds^2 = \left(\frac{X(p) - Y(q)}{p^2 q^4 - 2p^3 q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\tau \otimes d\tau + \left(\frac{q^2 X(p) + p^2 Y(q)}{p^2 q^4 - 2p^3 q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\tau \otimes d\sigma + \left(\frac{p^2 + q^2}{(p^2 q^2 - 2pq + 1)Y(q)} \right) dq \otimes dq \\ + \left(\frac{p^2 + q^2}{p^2 q^2 X(p) - 2pqX(p) + X(p)} \right) dp \otimes dp + \left(\frac{q^2 X(p) + p^2 Y(q)}{p^2 q^4 - 2p^3 q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\sigma \otimes d\tau + \left(\frac{q^4 X(p) - p^4 Y(q)}{p^2 q^4 - 2p^3 q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\sigma \otimes d\sigma$$

Coordinates

$$(x^\mu) = (\tau, q, p, \sigma)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\tau + \left(-\frac{\sqrt{2}\sqrt{p^2 + q^2}}{2(pq - 1)\sqrt{Y(q)}} \right) dq + \left(-\frac{\sqrt{2}p^2\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\sigma$$

$$n = \left(\frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\tau + \left(\frac{\sqrt{2}\sqrt{p^2 + q^2}}{2(pq - 1)\sqrt{Y(q)}} \right) dq + \left(-\frac{\sqrt{2}p^2\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\sigma$$

$$m = \left(-\frac{i\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\tau + \left(-\frac{\sqrt{2}\sqrt{p^2 + q^2}}{2(pq - 1)\sqrt{X(p)}} \right) dp + \left(-\frac{i\sqrt{2}q^2\sqrt{X(p)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\sigma$$

$$\bar{m} = \left(\frac{i\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\tau + \left(-\frac{\sqrt{2}\sqrt{p^2 + q^2}}{2(pq - 1)\sqrt{X(p)}} \right) dp + \left(\frac{i\sqrt{2}q^2\sqrt{X(p)}}{2\sqrt{p^2 + q^2}(pq - 1)} \right) d\sigma$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}pq^3}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}q^2}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \right) d\tau + \left(-\frac{\sqrt{2}pq\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} \right) dq + \left(\frac{\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \right) d\sigma$$

$$n = \left(-\frac{\sqrt{2}pq^3}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}q^2}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \right) d\tau + \left(\frac{\sqrt{2}pq\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} - \frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} \right) dq + \left(\frac{\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \right) d\sigma$$

$$m = \left(-\frac{i\sqrt{2}p^3q}{2\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{i\sqrt{2}p^2}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \right) d\tau + \left(-\frac{\sqrt{2}pq\sqrt{X(p)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2+q^2}} \right) dp + \left(-\frac{i\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{i\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \right) d\sigma$$

$$\bar{m} = \left(\frac{i\sqrt{2}p^3q}{2\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{i\sqrt{2}p^2}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \right) d\tau + \left(-\frac{\sqrt{2}pq\sqrt{X(p)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2+q^2}} \right) dp + \left(\frac{i\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{i\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \right) d\sigma$$

Directional Derivatives

$$D = -\frac{\sqrt{2}pq^3}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}q^2}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \partial_\tau - \frac{\sqrt{2}pq\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} \partial_q + \frac{\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \partial_\sigma$$

$$\Delta = -\frac{\sqrt{2}pq^3}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}q^2}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \partial_\tau + \frac{\sqrt{2}pq\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} - \frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} \partial_q + \frac{\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \partial_\sigma$$

$$\delta = -\frac{i\sqrt{2}p^3q}{2\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{i\sqrt{2}p^2}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \partial_\tau - \frac{\sqrt{2}pq\sqrt{X(p)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2+q^2}} \partial_p - \frac{i\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{i\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \partial_\sigma$$

$$\bar{\delta} = \frac{i\sqrt{2}p^3q}{2\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{i\sqrt{2}p^2}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \partial_\tau - \frac{\sqrt{2}pq\sqrt{X(p)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2+q^2}} \partial_p + \frac{i\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{i\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{X(p)}} \partial_\sigma$$

Spin Coefficients

$$\rho = -\frac{p^3 \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} + \frac{i p^2 q \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{i p \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{q \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{p^3 \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} + \frac{i p^2 q \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{i p \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{q \sqrt{Y(q)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\begin{aligned} \alpha = & \frac{\sqrt{2}\sqrt{p^2 + q^2}p^3q\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} + \frac{\sqrt{2}\sqrt{p^2 + q^2}pq^3\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} - \frac{\sqrt{2}\sqrt{p^2 + q^2}p^2q\sqrt{X(p)}}{2(p^4 + 2p^2q^2 + q^4)} + \frac{i\sqrt{2}\sqrt{p^2 + q^2}pq^2\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} - \frac{\sqrt{2}\sqrt{p^2 + q^2}q^3\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} \\ & - \frac{\sqrt{2}\sqrt{p^2 + q^2}p^2\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} - \frac{\sqrt{2}\sqrt{p^2 + q^2}q^2\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} + \frac{\sqrt{2}\sqrt{p^2 + q^2}p\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} - \frac{i\sqrt{2}\sqrt{p^2 + q^2}q\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} \end{aligned}$$

$$\begin{aligned} \beta = & -\frac{\sqrt{2}\sqrt{p^2 + q^2}p^3q\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} - \frac{\sqrt{2}\sqrt{p^2 + q^2}pq^3\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} + \frac{\sqrt{2}\sqrt{p^2 + q^2}p^2q\sqrt{X(p)}}{2(p^4 + 2p^2q^2 + q^4)} - \frac{i\sqrt{2}\sqrt{p^2 + q^2}pq^2\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} \\ & + \frac{\sqrt{2}\sqrt{p^2 + q^2}q^3\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} + \frac{\sqrt{2}\sqrt{p^2 + q^2}p^2\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} + \frac{\sqrt{2}\sqrt{p^2 + q^2}q^2\frac{\partial}{\partial p}X(p)}{8(p^4 + 2p^2q^2 + q^4)\sqrt{X(p)}} - \frac{\sqrt{2}\sqrt{p^2 + q^2}p\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} + \frac{i\sqrt{2}\sqrt{p^2 + q^2}q\sqrt{X(p)}}{4(p^4 + 2p^2q^2 + q^4)} \end{aligned}$$

$$\tau = -\frac{i pq^2 \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{q^3 \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{p \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} + \frac{i q \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}}$$

$$\pi = \frac{i pq^2 \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} + \frac{q^3 \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} + \frac{p \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}} - \frac{i q \sqrt{X(p)}}{(\sqrt{2}p^2 + \sqrt{2}q^2)\sqrt{p^2 + q^2}}$$

$$\begin{aligned}
\epsilon = & -\frac{\sqrt{2}\sqrt{p^2+q^2}p^3q\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} - \frac{\sqrt{2}\sqrt{p^2+q^2}pq^3\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^3\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^2q\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} + \frac{\sqrt{2}\sqrt{p^2+q^2}pq^2\sqrt{Y(q)}}{2(p^4+2p^2q^2+q^4)} \\
& + \frac{\sqrt{2}\sqrt{p^2+q^2}p^2\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} + \frac{\sqrt{2}\sqrt{p^2+q^2}q^2\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} - \frac{i\sqrt{2}\sqrt{p^2+q^2}p\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} - \frac{\sqrt{2}\sqrt{p^2+q^2}q\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} \\
\gamma = & -\frac{\sqrt{2}\sqrt{p^2+q^2}p^3q\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} - \frac{\sqrt{2}\sqrt{p^2+q^2}pq^3\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^3\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^2q\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} + \frac{\sqrt{2}\sqrt{p^2+q^2}pq^2\sqrt{Y(q)}}{2(p^4+2p^2q^2+q^4)} \\
& + \frac{\sqrt{2}\sqrt{p^2+q^2}p^2\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} + \frac{\sqrt{2}\sqrt{p^2+q^2}q^2\frac{\partial}{\partial q}Y(q)}{8(p^4+2p^2q^2+q^4)\sqrt{Y(q)}} - \frac{i\sqrt{2}\sqrt{p^2+q^2}p\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)} - \frac{\sqrt{2}\sqrt{p^2+q^2}q\sqrt{Y(q)}}{4(p^4+2p^2q^2+q^4)}
\end{aligned}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = & -\frac{p^4q^2\frac{\partial^2}{(\partial p)^2}X(p)}{8(p^4+2p^2q^2+q^4)} - \frac{p^2q^4\frac{\partial^2}{(\partial p)^2}X(p)}{8(p^4+2p^2q^2+q^4)} + \frac{p^4q^2\frac{\partial^2}{(\partial q)^2}Y(q)}{8(p^4+2p^2q^2+q^4)} + \frac{p^2q^4\frac{\partial^2}{(\partial q)^2}Y(q)}{8(p^4+2p^2q^2+q^4)} + \frac{3p^3q^2\frac{\partial}{\partial p}X(p)}{4(p^4+2p^2q^2+q^4)} + \frac{pq^4\frac{\partial}{\partial p}X(p)}{4(p^4+2p^2q^2+q^4)} - \frac{p^4q\frac{\partial}{\partial q}Y(q)}{4(p^4+2p^2q^2+q^4)} \\
& - \frac{3p^2q^3\frac{\partial}{\partial q}Y(q)}{4(p^4+2p^2q^2+q^4)} - \frac{3p^2q^2X(p)}{2(p^4+2p^2q^2+q^4)} + \frac{3p^2q^2Y(q)}{2(p^4+2p^2q^2+q^4)} + \frac{p^3q\frac{\partial^2}{(\partial p)^2}X(p)}{4(p^4+2p^2q^2+q^4)} + \frac{pq^3\frac{\partial^2}{(\partial p)^2}X(p)}{4(p^4+2p^2q^2+q^4)} - \frac{p^3q\frac{\partial^2}{(\partial q)^2}Y(q)}{4(p^4+2p^2q^2+q^4)} - \frac{pq^3\frac{\partial^2}{(\partial q)^2}Y(q)}{4(p^4+2p^2q^2+q^4)} \\
& - \frac{5p^2q\frac{\partial}{\partial p}X(p)}{4(p^4+2p^2q^2+q^4)} - \frac{q^3\frac{\partial}{\partial p}X(p)}{4(p^4+2p^2q^2+q^4)} + \frac{p^3\frac{\partial}{\partial q}Y(q)}{4(p^4+2p^2q^2+q^4)} + \frac{5pq^2\frac{\partial}{\partial q}Y(q)}{4(p^4+2p^2q^2+q^4)} + \frac{2pqX(p)}{p^4+2p^2q^2+q^4} - \frac{2pqY(q)}{p^4+2p^2q^2+q^4} - \frac{p^2\frac{\partial^2}{(\partial p)^2}X(p)}{8(p^4+2p^2q^2+q^4)} \\
& - \frac{q^2\frac{\partial^2}{(\partial p)^2}X(p)}{8(p^4+2p^2q^2+q^4)} + \frac{p^2\frac{\partial^2}{(\partial q)^2}Y(q)}{8(p^4+2p^2q^2+q^4)} + \frac{q^2\frac{\partial^2}{(\partial q)^2}Y(q)}{8(p^4+2p^2q^2+q^4)} + \frac{p\frac{\partial}{\partial p}X(p)}{2(p^4+2p^2q^2+q^4)} - \frac{q\frac{\partial}{\partial q}Y(q)}{2(p^4+2p^2q^2+q^4)} - \frac{X(p)}{2(p^4+2p^2q^2+q^4)} + \frac{Y(q)}{2(p^4+2p^2q^2+q^4)}
\end{aligned}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\begin{aligned} \Lambda = & -\frac{p^2 q^2 \frac{\partial^2}{(\partial p)^2} X(p)}{24(p^2 + q^2)} - \frac{p^2 q^2 \frac{\partial^2}{(\partial q)^2} Y(q)}{24(p^2 + q^2)} + \frac{pq^2 \frac{\partial}{\partial p} X(p)}{4(p^2 + q^2)} + \frac{p^2 q \frac{\partial}{\partial q} Y(q)}{4(p^2 + q^2)} - \frac{q^2 X(p)}{2(p^2 + q^2)} - \frac{p^2 Y(q)}{2(p^2 + q^2)} \\ & + \frac{pq \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^2 + q^2)} + \frac{pq \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^2 + q^2)} - \frac{q \frac{\partial}{\partial p} X(p)}{4(p^2 + q^2)} - \frac{p \frac{\partial}{\partial q} Y(q)}{4(p^2 + q^2)} - \frac{\frac{\partial^2}{(\partial p)^2} X(p)}{24(p^2 + q^2)} - \frac{\frac{\partial^2}{(\partial q)^2} Y(q)}{24(p^2 + q^2)} \end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned} \Psi_2 = & \frac{p^6 q^2 \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^4 q^4 \frac{\partial^2}{(\partial p)^2} X(p)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^2 q^6 \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^6 q^2 \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^4 q^4 \frac{\partial^2}{(\partial q)^2} Y(q)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\ & + \frac{p^2 q^6 \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^5 q^2 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{i p^4 q^3 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^3 q^4 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{i p^2 q^5 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\ & - \frac{i p^5 q^2 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^3 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{i p^3 q^4 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^5 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^4 q^2 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} \\ & - \frac{2i p^3 q^3 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{p^2 q^4 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{p^4 q^2 Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{2i p^3 q^3 Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{p^2 q^4 Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} \\ & - \frac{p^5 q \frac{\partial^2}{(\partial p)^2} X(p)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^3 q^3 \frac{\partial^2}{(\partial p)^2} X(p)}{3(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{pq^5 \frac{\partial^2}{(\partial p)^2} X(p)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^5 q \frac{\partial^2}{(\partial q)^2} Y(q)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^3 q^3 \frac{\partial^2}{(\partial q)^2} Y(q)}{3(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\ & - \frac{pq^5 \frac{\partial^2}{(\partial q)^2} Y(q)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^4 q \frac{\partial}{\partial p} X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{i p^3 q^2 \frac{\partial}{\partial p} X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{p^2 q^3 \frac{\partial}{\partial p} X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{i pq^4 \frac{\partial}{\partial p} X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} \\ & + \frac{i p^4 q \frac{\partial}{\partial q} Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{p^3 q^2 \frac{\partial}{\partial q} Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{i p^2 q^3 \frac{\partial}{\partial q} Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{pq^4 \frac{\partial}{\partial q} Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{2 p^3 q X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} \\ & + \frac{4i p^2 q^2 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{2 pq^3 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \frac{2 p^3 q Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{4i p^2 q^2 Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{2 pq^3 Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \dots \end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"